3a) Role of Query Decomposition in Distributed Query Processing

Answer:

Role of Query Decomposition:

Query decomposition is the first phase of distributed query processing. Its main role is to convert a high-level SQL query into an intermediate relational algebra form and then break it into simpler subqueries that can be executed efficiently on multiple distributed sites.

Key roles include:

1. Translation: Converts a SQL query into relational algebra.
2. Normalization: Rewrites the query to eliminate redundancy and simplify expressions.
3. Analysis: Checks for correctness, verifies data sources, and ensures security and authorization.
4. Simplification: Applies transformations and optimizations to improve performance.
5. Fragmentation Mapping: Maps the query to corresponding data fragments stored at different sites.

Example:

Consider a high-level SQL query:

```
SELECT E.Name, D.DeptName
FROM Employee E, Department D
WHERE E.DeptID = D.DeptID AND D.Location = 'Kathmandu';
```

Assume:

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- Employee table is stored at Site A
- Department table is stored at Site B

Decomposition Process:

1. Translation to Relational Algebra:

PROJECTION_{E.Name, D.DeptName} (SELECTION_{D.Location='Kathmandu'}(Department) JOIN Employee)

2. Decompose into Subqueries:

- Subquery at Site B:

  SELECT DeptID, DeptName FROM Department WHERE Location = 'Kathmandu';

- Subquery at Site A:

  Join result of above with Employee data based on DeptID.

3. Optimization:

   - Transfer only required rows from Site B to Site A.
   - Perform join at the site with fewer rows (Site A, assuming Employee table is smaller after filtering).

Final Execution Strategy:

- Fetch departments located in Kathmandu from Site B.

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- Send this result to Site A.

- Join with Employee table at Site A.

- Return the final result.

Benefits:

- Reduces communication cost.

- Allows parallel execution across multiple sites.

- Improves performance and scalability of distributed databases.